



FPDAF (Federated Personalized Drift Alerting Framework): CLINICAL AI EARLY-WARNING WORKSTATION

Explainable Federated Learning for Sepsis CDSS with Personalized Drift Adaptation

MSME Idea Hackathon 6.0 Proposal

Sub-Theme 6: Frontier Technologies

Student Innovators: Sheela Akshar Sakhi, Hasini Reddy M, Kousik Sarma Lakkaraju,
Haswitha K, V. Chakravarthy

Mentors: Dr. Ramya G. R. (Lead Mentor) & Dr. Vandhana S. (Co-Mentor)
Department of CSE, Amrita School of Computing, Amrita Vishwa Vidyapeetham, Coimbatore

[Risk] Sepsis Pathological Threat

Sepsis is an extreme, systemic immune response to infection, causing rapid organ dysfunction.

- ▶ **Septic Shock Crash:** Multi-organ failure and blood pressure drop.
- ▶ **The Time Challenge:** Survival drops by **8% per hour** of delayed administration.
- ▶ **Diagnosis Bottleneck:** Initial symptoms are masked, mimicking minor infections.

[Metrics] Critical Scale in India

Sepsis is a leading clinical bottleneck:

- ▶ **2.9 to 3.0 Million Deaths:** Estimated annually across India (accounting for **30%** of total deaths).
- ▶ **ICU Mortality Rates:** Ranges from **34% to over 50%** inside Indian intensive care units.

Source: [George Institute for Global Health \(2020\)](#) | ISCCM

Clinical Data Privacy Obstacle

Traditional centralized AI requires uploading highly sensitive Patient Health Information (PHI) to external clouds, violating strict data-sovereignty regulations such as HIPAA and the Indian DPDPA (Section 8).

The Objective: Decentralized Clinical Early Warning

FPDAF (Federated Personalized Drift Alerting Framework) is an intelligent, secure Clinical Decision Support System (CDSS) running locally in intensive care units. Using Federated Learning, hospital clients train a shared model collaboratively while keeping patient charts local.

1. Real-time Telemetry

Continuously segments and scales sliding 24h patient streams (vitals & labs) on local server nodes.

2. Secure Federated Core

Hospital firewalls remain closed. Clients swap only weight tensors (FedAvg/FedProx) with the coordinate server.

3. Bedside Warnings

Generates stable vs. critical notifications along with explainable temporal heatmap visualizations.

Uniqueness of the Innovation (Key Technical Novelties)

1. Multi-Head Temporal XAI

Embeds 4-head query-key temporal self-attentions providing explainable heatmap insights into exact sequence hours (e.g. SpO₂ drop at hour 14) driving risk alerts.

2. CUSUM Concept Drift Auditing

An online Cumulative Sum (AD-CUSUM) neural monitor tracks validation loss residuals at each hospital site to flag patient demographic shifts in real-time.

3. Dynamic Dual-Phase (DDP) Attention

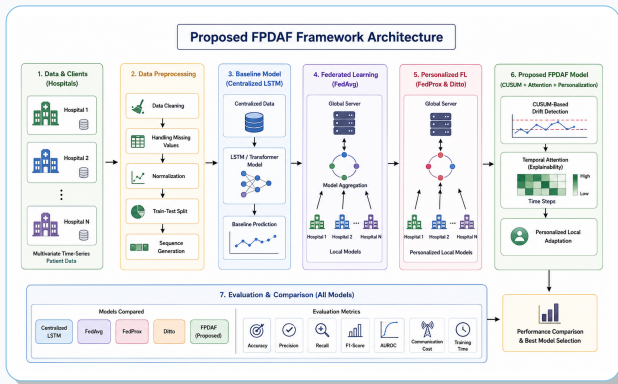
Integrates 2D cross-attention ($\alpha_t \otimes \beta_v$) to capture simultaneous sequence time steps AND multi-modal vital co-dependencies (e.g., HR vs Blood Pressure).

4. Client-Side Selective Personalization

Freezes the shared deep LSTM backbone upon drift alerts, fine-tuning only local classifier heads:

- ▶ **38% Bandwidth Savings** (1.52 GB vs 2.48 GB).
- ▶ **Rapid Recalibration:** Adapts to hospital profiles in **6 minutes**.

Flowchart of the Proposed Solution



System Dataflow: Local preprocessed vital arrays flow through the shared BiLSTM feature extractor, attention query layers, and personalized heads, controlled by local CUSUM triggers.

Validated Experimental Performance

Evaluated across multi-center PhysioNet ICU cohorts:

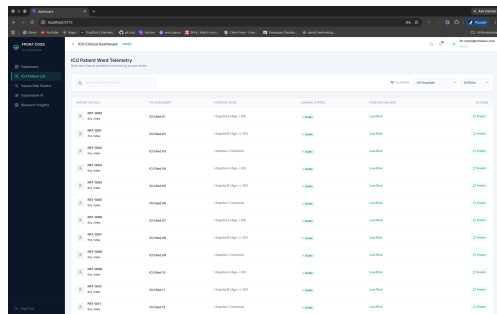
- ▶ **86.51% Test Accuracy:** Outperforms FedAvg (75.94%) and FedProx (78.85%) under non-IID data splits.
- ▶ **0.8214 AUROC Score:** Strong discrimination for early sepsis crash prediction.
- ▶ **38% Bandwidth Reduction:** 1.52 GB network usage vs 2.48 GB in standard FL.

Software & Database Workstation

Full-stack bedside workstation operational:

- ▶ **FastAPI Backend & SQLite Warehousing:** Decodes raw vitals using `scaler.pkl`.
- ▶ **React Vite Bedside UI:** Features Doctor & Admin portals

Bedside Workstation Telemetry Interface



The screenshot displays a web application titled "ICU Patient Ward Telemetry". It features a sidebar with navigation options like "ICU Patient Ward Telemetry", "ICU Patient Ward Telemetry", "ICU Patient Ward Telemetry", and "ICU Patient Ward Telemetry". The main content area shows a table with columns: "PATIENT ID", "ICU FLOOR", "HOSPITAL WARD", "CURRENT STATUS", "CURRENT RISK", and "ACTION". The table lists 10 patients with their respective IDs, floors, wards, and current status (e.g., "Hospitals Edge - 100", "Hospitals Edge - 100", "Hospitals Edge - 100").

PATIENT ID	ICU FLOOR	HOSPITAL WARD	CURRENT STATUS	CURRENT RISK	ACTION
1001-1001	ICU Floor 01	Hospitals Edge - 100	100%	Low Risk	100%
1001-1002	ICU Floor 02	Hospitals Edge - 100	100%	Low Risk	100%
1001-1003	ICU Floor 03	Hospitals Edge - 100	100%	Low Risk	100%
1001-1004	ICU Floor 04	Hospitals Edge - 100	100%	Low Risk	100%
1001-1005	ICU Floor 05	Hospitals Edge - 100	100%	Low Risk	100%
1001-1006	ICU Floor 06	Hospitals Edge - 100	100%	Low Risk	100%
1001-1007	ICU Floor 07	Hospitals Edge - 100	100%	Low Risk	100%
1001-1008	ICU Floor 08	Hospitals Edge - 100	100%	Low Risk	100%
1001-1009	ICU Floor 09	Hospitals Edge - 100	100%	Low Risk	100%
1001-1010	ICU Floor 10	Hospitals Edge - 100	100%	Low Risk	100%

Bedside Doctor Portal: Real-time vital timelines, risk dials, and temporal attention saliency heatmaps.

Competitive Matrix: Comparison of Key Capabilities

Feature	FPDAF (Ours)	Epic System	Philips Monitor	Google I
Federated Privacy	Yes	No (Cloud)	No (Local)	No (Cloud)
Concept Drift Auditing	Yes	No	No	No
Temporal XAI Heatmaps	Yes	No	No	Yes
Local Customization (CSSP)	Yes	No	No	No
Edge-AI Deployment	Yes	Partial	Yes	No
DPDPA-Ready Compliance	Yes	Partial	Partial	Partial

Strategic Advantage

While big tech offers cloud AI and hardware brands offer dumb local alarms, FPDAF is the only system offering:

- ▶ **Privacy-first training**
- ▶ **Bedside temporal explainability**
- ▶ **Local adaptation to prevent medical drift.**

* Based on publicly available product specifications and literature review as of Q2.

† Source: Indian Healthcare CDSS Market Analysis Report (2025).

Indian MSME Hardware Integration

Unlike pure software startups, FPDAF (Federated Personalized Drift Alerting Framework) is proposed as an **integrated medical hardware product**:

- ▶ **Physical Node Box:** A bedside workstation cabinet containing custom local display, visual/audio alarms, and network interface.
- ▶ **Local Assembly Partner:** Discussions initiated with local embedded electronics and medical device MSMEs in Coimbatore for PCB assembly and edge hardware enclosure fabrication.

Industrial Applications

- ▶ **Hospital ICU Deployment:** Integrates with patient monitors via lightweight APIs to run as an intelligent local CDSS layer.
- ▶ **OEM Licensing:** Licensing the lightweight, PyTorch-based early-warning algorithm to patient monitoring equipment manufacturers (e.g. Philips, GE Healthcare).

TAM/SAM/SOM Sizing

Detailed market mapping highlights scaling capacity:

- ▶ **Estimated TAM: Rs. 1,000 Crores** (India's Clinical Decision Support System market)[†].
- ▶ **Estimated SAM: Rs. 375 Crores** (Corporate/private hospital ICU networks representing 2,50,000 private ICU beds).
- ▶ **Estimated SOM: Rs. 40 Crores** (Regional hospital networks targetable in Years 1–2 representing 12 chains).

Government Adoption

Designed for scalable deployment across public healthcare infrastructure:

- ▶ **Potential Users:** AIIMS, State Medical Colleges, Government Hospitals, District Hospitals, Corporate Chains, Tele-ICU

Commercialization & Scaling Strategy

We scale through a 5-Phase commercialization strategy:

1. **Phase 1:** Pilot hospital validation.
2. **Phase 2:** Regional network deployment.
3. **Phase 3:** Government procurement and public hospital deployment.
4. **Phase 4:** Scaling to private hospital networks.
5. **Phase 5:** Global export scaling.

Proposed Budget Justification (Total: Rs. 15.0 Lakhs)

Budget Request Under MSME Idea Hackathon 6.0 Guidelines (Maximum Grant):

We request Rs. 15 Lakhs to fund the transition of our working prototype into physical clinical pilot installations:

1. Technology Costs

Rs. 10.0 Lakhs

- ▶ **Hardware & Assembly (3.6L):** 3 Edge-AI Jetson nodes, PCB assembly, medical touch displays, workstation enclosures with Coimbatore MSMEs.
- ▶ **Data Annotation (3.4L):** Expert clinician curation & vital stream simulation.
- ▶ **Security Audits (3.0L):** Penetration testing & DPDPA audits.

2. Clinical Mentorship

Rs. 3.0 Lakhs

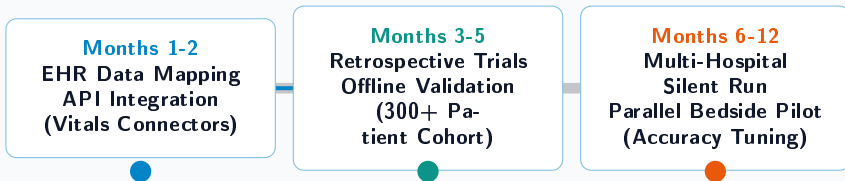
- ▶ ICU clinical validation guidance (1.8L).
- ▶ Incubation facility & testing lab support (1.2L).

3. Field Travel & Trials

Rs. 2.0 Lakhs

- ▶ Bedside node hardware setups (1.0L).
- ▶ Multi-site ICU telemetry audits (1.0L).

Project Development Roadmap (12 Months)



Commercialization Goal (Post 12 Months)

Acquire CDSS medical software safety certifications, apply for patent filings on the CUSUM trigger mechanisms, and deploy commercial B2B SaaS nodes across Indian ICU networks.

Why FPDAF (Federated Personalized Drift Alerting Framework) is the Perfect Fit for MSME Funding

[Impact] High Social & Medical Impact

Sepsis is the leading cause of ICU mortality. Early bedside warning directly:

- ▶ Improves patient clinical outcomes through early clinical fluid and antibiotic intervention.
- ▶ Reduces ICU average length of stay (ALoS), helping to potentially reduce ICU treatment costs.

[Alignment] National Mission Alignment

- ▶ **Aatmanirbhar Bharat:** Completely local design, reducing dependency on imported clinical systems.
- ▶ **Viksit Bharat @2047:** Empowers rural and semi-urban health centers with low-cost clinical intelligence.
- ▶ **High Feasibility:** Functional prototype developed, compiled, and validated on actual public EHR cohorts.

Thank You

Explainable Healthcare AI to Save Lives at the Bedside

MSME Idea Hackathon 6.0

Theme: **Frontier Technologies (Sub-Theme 6)**

Anonymous Project Submission

Host Institute Code: MSME-HI-REDACTED

Questions & Discussion